

Managerial Accounting — Practice Questions

Part 1 is auto-generated per lecture (quick recap + one exam-style question testing the trickiest concept covered). Part 2 is extracted directly from the course's own question banks — verbatim, not LLM-regenerated, so treat those as authoritative.

Part 1: Lecture-by-Lecture Quick Practice

Lecture 01 (2026-04-26)

Topic: Research vs. Development Distinction

Logic: Research is treated as a cost center because it lacks a guaranteed future benefit, while Development is treated as an asset because it is close to finding a solution and has a clear future benefit.

Practice Question: A company spends \$200,000 on a project with no guaranteed future benefit and \$100,000 on a project that is close to finding a solution and has a clear future benefit. Classify each expenditure as a "cost center" or an "asset."

Solution: \$200,000 is a cost center; \$100,000 is an asset.

Why: This tests the "guaranteed future benefit" distinction for classification.

Lecture 02 (2026-05-03)

Topic: Asset vs. Liability Identification

Logic: Stand at a point in time and look forward: if a resource provides a "future benefit," it is an Asset; if it creates a "future obligation," it is a Liability.

Practice Question: A company receives \$10,000 from a student for a course to be taught next semester. The company also provides a \$5,000 loan to a supplier. Classify both items as either an Asset or a Liability.

Solution: 1. Student payment: Liability (it is "pre-received income" which represents a "future obligation" to provide service). 2. Loan to supplier: Asset (it is a "loan given" which provides a "future benefit").

Why: This tests the "future benefit" vs. "future obligation" rule applied to "pre-received income" and "loans given."

Lecture 03 (2026-05-10)

Topic: Capital Lease and Goodwill Valuation

Logic: For a capital lease, the Lease Liability is recorded as the book value of the asset, not the amount of the rent paid. Goodwill is calculated as the purchase price minus the fair value of the company at the time of acquisition.

Practice Question: A company enters into a capital lease for a piece of land with a book value of 500 units; the annual rent for the lease is 50 units. In a separate transaction, the company acquires a neighboring business with a fair value of 1,000 units for a total purchase price of 1,200 units. Calculate the Lease Liability and the Goodwill.

Solution: 1. Lease Liability: 500 (The liability equals the book value of the asset, not the rent amount). 2. Goodwill: 200 (1,200 purchase price - 1,000 fair value).

Why: This tests the distinction between rent payments and asset book value in lease accounting, and the specific "premium over fair value" calculation for goodwill.

Lecture 04 (2026-05-17)

Topic: Prepaid Interest vs. Expense Classification

Logic: If the "time to pay" (the period it applies to) has not yet arrived, the payment is a future benefit and is classified as an asset; if the benefit is "used" in the current period, it is an expense.

Practice Question: A company pays 8,000 for interest where the "time to pay" has not arrived. In the following period, this interest is "used" in the current period. Classify both transactions.

Solution: 1. Payment where time has not arrived: Asset (Prepaid Interest). 2. Amount used in current period: Expense (Interest Expense).

Why: This tests the "timing" rule to distinguish between a future benefit (Asset) and a consumed benefit (Expense).

Lecture 05 (2026-05-24)

Topic: Rent Classification and Expense Rules

Logic: Assets and expenses are on the LHS, while liabilities and income are on the RHS. "Rent Outstanding" is a liability (RHS); because it was already recognized as an expense in a prior period, its payment is not recorded as an expense in the current period.

Practice Question: A company pays ₹5,000 for "Rent Outstanding" from the previous year. How is this transaction recorded, and is it recorded as an expense?

Solution: Debit: Rent Outstanding (₹5,000); Credit: Cash (₹5,000). It is not recorded as an expense because it was already recognized as an expense in a prior period.

Why: This tests the distinction between a liability reduction and a current-period expense.

Lecture 06 (2026-05-31)

Topic: Journal Entry Classification and Specific Account Usage

Logic: Assets are recorded on the Left-Hand Side (LHS) and increase on Debit; Liabilities and Equity are on the Right-Hand Side (RHS) and increase on Credit. Specific terms must be used: "Drawings" for personal withdrawals (rather than "Capital") and "Machinery" for tangible assets (rather than "Purchase").

Practice Question: An owner withdraws ₹10,000 for personal use, and the company purchases a piece of machinery for ₹30,000. Provide the correct Debit and Credit entries for both transactions.

Solution: 1. Withdrawal: Debit: Drawing account Dr. 10,000; Credit: To Bank account 10,000 2. Machinery: Debit: Machinery account Dr. 30,000; Credit: To Bank account 30,000

Why: This tests the requirement to use specific names for personal withdrawals ("Drawings") and tangible assets ("Machinery") rather than generic terms.

Lecture 07 (2026-06-07)

Topic: Profit and Loss Item Classification

Logic: Dividends are not an expense; they are a "sharing of profit" or an appropriation of profit. Pre-received income is classified as a liability because it represents a "future obligation."

Practice Question: A company receives \\$10,000 in dividends and \\$5,000 in "pre-received income." Classify each as Income, Expense, or Liability.

Solution: Dividend: Income. Pre-received Income: Liability.

Why: This tests the "tricky" distinction between appropriations of profit and future obligations.

Lecture 08 (2026-06-14)

Topic: Prepaid and Outstanding Expense Adjustments

Logic: Prepaid expenses are not "current" expenses and must be removed from the current period's expense and recorded as an asset. Outstanding expenses have occurred but are not yet paid; they must be included in the current period's costs (reducing profit) and recorded as a liability.

Practice Question: A company has a reported expense of 10,000 for Rent, of which 4,000 is Prepaid. There is also an Outstanding Salary of 3,000. Calculate the amount to be included in the current period's costs and identify the Asset and Liability values.

Solution: Current period's costs: 9,000 ($10,000 - 4,000 + 3,000$); Asset: 4,000; Liability: 3,000.

Why: This tests the distinction between items that are not "current" expenses (prepaid) and items that have occurred but are not yet paid (outstanding).

Lecture 09 (2026-07-05)

Topic: Return on Investment (ROI) Calculation

Logic: Profit is a figure recorded over a continuous period, while Investment is a "point-in-time" figure from the Balance Sheet. To make these quantities comparable for a ratio, the "Standard Solution" is to use the average of the investment: $(\text{Opening Investment} + \text{Closing Investment}) / 2$.

Practice Question: A company reports a Profit of \$100,000 for the fiscal year. The investment at the start of the period was \$1,000,000 and the investment at the end of the period was \$1,200,000. Calculate the Return on Investment (ROI).

Solution: 1. Calculate the average investment: $(\$1,000,000 + \$1,200,000) / 2 = \$1,100,000$. 2. Calculate the ROI: $(\$100,000 / \$1,100,000) \times 100 = 9.09\%$.

Why: This tests the "Standard Solution" for resolving the comparison mismatch between period-based profit and point-in-time investment figures.

Lecture 10 (2026-07-12)

Topic: Diluted Earnings Per Share (EPS)

Logic: Basic EPS is calculated using current equity shares, whereas Diluted EPS is a more conservative measure because it includes all shares that can be converted into equity in the future, such as convertible

bonds, warrants, options, and ESOPs.

Practice Question: A company reports a Net Profit of \$1,000,000 and pays \$100,000 in preference dividends. There are 100,000 current equity shares. Additionally, the company has convertible bonds and warrants that could be converted into 20,000 additional shares. Calculate both the Basic EPS and the Diluted EPS.

Solution: Basic EPS: $(\$1,000,000 - \$100,000) / 100,000 = \$9.00$ Diluted EPS: $(\$1,000,000 - \$100,000) / (100,000 + 20,000) = \7.50

Why: This question tests the requirement to include potential equity-convertible instruments (convertible bonds, warrants) in the denominator to provide a more conservative Diluted EPS.

Lecture 11 (2026-07-19)

Topic: Sunk Costs and Opportunity Costs

Logic: Sunk costs are costs already incurred and cannot be recovered, making them irrelevant for future decision-making. Opportunity costs represent the value of the next best alternative foregone and serve as the "minimum requirement" a decision must meet to be justified.

Practice Question: A company spent 40,000 on an old software system. They are now considering purchasing a new version for 100,000. Is the 60,000 difference between the old and new systems' costs considered an opportunity cost?

Solution: No, the 60,000 is a price difference. The 40,000 spent on the old system is the sunk cost because it cannot be recovered.

Why: This tests the distinction between a sunk cost and a price difference, ensuring the student does not incorrectly classify a price delta as an opportunity cost.

Lecture 12 (2026-07-26)

Topic: CVP Analysis and Break-Even Point

Logic: Contribution margin is the amount remaining from sales revenue after variable costs are deducted. To find how many more units must be sold to cover fixed costs, divide the total fixed costs by the contribution margin per unit and subtract the number of units already sold.

Practice Question: A company currently sells 8,000 units and has a loss of 7,000. The fixed costs are 35,000 and the contribution margin per unit is 3.50. How many more units must be sold to cover the fixed costs?

Solution: 1. Calculate total units needed to cover fixed costs: $35,000 / 3.50 = 10,000$ total units. 2. Calculate additional units: $10,000 - 8,000 = 2,000$ units.

Why: This tests the calculation of "additional" units required to reach the break-even point rather than just the total number of units.

Lecture 13 (2026-08-02)

Topic: Drop Product Decision Analysis

Logic: A decision to drop a product must be based on the impact on the contribution to the company's profit. If the reduction in fixed costs is less than the loss in contribution margin from dropping the product,

the product should not be dropped.

Practice Question: A company is considering dropping its "Mud and Snow" tire line. - With the line: Sales = \$180,000; Fixed Overhead = \$30,000; Contribution Margin = \$7,000. - Without the line: Sales = \$155,000; Fixed Overhead = \$28,000; Contribution Margin = \$5,000. Should the company drop the product?

Solution: No. By dropping the product, sales drop by \$25,000, but fixed costs only decrease by \$2,000; therefore, the contribution to the company's profit would drop from \$7,000 to \$5,000.

Why: This tests the ability to identify the "gap" between the loss in contribution and the reduction in fixed costs.

Lecture 14 (2026-08-09)

Topic: Margin of Safety (MoS) and Profit Calculation

Logic: The Margin of Safety (MoS) is the amount of sales that exceeds the break-even point. Since only this "extra" amount sold beyond break-even contributes to profit, profit is calculated by multiplying the MoS by the Profit Volume (P/V) ratio.

Practice Question: A company records a Margin of Safety (MoS) of 100,000 and a Profit Volume (P/V) Ratio of 40%. Calculate the profit.

Solution: $\$100,000 \times 0.40 = 40,000\$$

Why: This tests the specific logic that only the portion of sales exceeding the break-even point (the "extra" amount) contributes to profit.

Lecture 15 (2026-08-23)

Topic: Standard Costing: Material and Labor Variances

Logic: Standard Costing compares actual costs against predetermined "standards" to determine if a project is being performed well. By breaking total costs into components—such as Price/Usage for materials or Rate/Efficiency for labor—management can identify if a variance is caused by the price paid or the quantity used.

Practice Question: A company has a Standard Quantity (SQ) of 2,500 kg at a Standard Price (SP) of \$12/kg. The Actual Quantity (AQ) used was 2,200 kg at an Actual Price (AP) of \$13/kg. Calculate the Material Cost Variance (MCV), Material Price Variance (MPV), and Material Usage Variance (MUV). State if each is Favorable (F) or Adverse (A).

Solution: 1. **MCV:** $\$(2,500 \times 12) - (2,200 \times 13) = 30,000 - 28,600 = 1,400\$$ (F) 2. **MPV:** $\$2,200 \times (12 - 13) = -2,200\$$ (A) 3. **MUV:** $\$12 \times (2,500 - 2,200) = 3,600\$$ (F)

Why: This tests the ability to distinguish between price and usage as independent components of the total variance.

Lecture 16 (2026-08-30)

Topic: Cash Flow from Operations (Indirect Method)

Logic: The Indirect Method identifies Cash Flow from Operations (CFO) by starting with a profit figure (PAT) and "removing" non-cash items and non-operating items. Non-cash expenses (Depreciation) and

non-operating costs (Finance Cost) are added back because they were subtracted in the P&L, while non-operating gains are subtracted because they were added to the P&L.

Practice Question: A company reports a Profit After Tax (PAT) of \$200,000. During the period, it recorded Depreciation of \$20,000, a Finance Cost of \$10,000, a Gain on Sale of Assets of \$5,000, and an increase in Trade Receivables of \$15,000. Calculate the Cash Flow from Operations (CFO).

Solution: 1. Start with PAT: \$200,000 2. Add Depreciation: +\$20,000 (Non-cash expense) 3. Add Finance Cost: +\$10,000 (Non-operating cost; added back to "remove" from calculation) 4. Subtract Gain on Sale: -\$5,000 (Non-operating gain; subtracted to "remove" from calculation) 5. Subtract Increase in Trade Receivables: -\$15,000 (Increase in asset means cash is "locked" and not available) 6. Total CFO: \$210,000

Why: This tests the student's ability to distinguish between different reasons for "adding back" items (non-cash vs. non-operating costs) and correctly identifying the impact of working capital changes.

Part 2: Existing Course Question Banks (from shared materials)

Financial_Ratio_Analysis_20_Practice_Questions_with_Solutions.docx

FINANCIAL RATIO ANALYSIS 20 CLASSROOM PRACTICE QUESTIONS WITH SOLUTIONS

Financial and Management Accounting | Discussion Practice

Instructions: Attempt each question before discussing the solution. Use 365 days for turnover-period calculations unless otherwise stated.

Question 1 — Liquidity Ratios

A company has current assets of ₹12,00,000 and current liabilities of ₹6,00,000. Calculate the current ratio and comment on liquidity.

Solution: Current Ratio = Current Assets ÷ Current Liabilities = ₹12,00,000 ÷ ₹6,00,000 = 2:1. The company has ₹2 of current assets for every ₹1 of current liabilities. This matches the conventional benchmark of 2:1 and indicates comfortable short-term liquidity.

Question 2 — Liquidity Ratios

Current assets are ₹9,00,000, including inventory of ₹2,50,000 and prepaid expenses of ₹50,000. Current liabilities are ₹4,00,000. Calculate the quick ratio.

Solution: Quick Assets = Current Assets – Inventory – Prepaid Expenses = ₹9,00,000 – ₹2,50,000 – ₹50,000 = ₹6,00,000. Quick Ratio = ₹6,00,000 ÷ ₹4,00,000 = 1.5:1.

Question 3 — Liquidity Ratios

Cash in hand is ₹80,000, bank balance ₹1,20,000, short-term investments ₹2,00,000 and current liabilities ₹8,00,000. Calculate the absolute liquid ratio.

Solution: Absolute Liquid Assets = Cash + Bank + Short-term Investments = ₹80,000 + ₹1,20,000 + ₹2,00,000 = ₹4,00,000. Absolute Liquid Ratio = ₹4,00,000 ÷ ₹8,00,000 = 0.5:1.

Question 4 — Liquidity Ratios

A firm's current ratio is 2.5:1 and current liabilities are ₹4,00,000. Find current assets. If inventory is ₹3,00,000 and prepaid expenses ₹50,000, calculate the quick ratio.

Solution: Current Assets = $2.5 \times ₹4,00,000 = ₹10,00,000$. Quick Assets = $₹10,00,000 - ₹3,00,000 - ₹50,000 = ₹6,50,000$. Quick Ratio = $₹6,50,000 \div ₹4,00,000 = 1.625:1$, or approximately 1.63:1.

Question 5 — Liquidity Ratios

A company has cash ₹1,00,000, debtors ₹4,00,000, inventory ₹5,00,000, prepaid expenses ₹50,000 and current investments ₹1,50,000. Current liabilities are ₹5,00,000. Calculate current, quick and absolute liquid ratios.

Solution: Current Assets = ₹12,00,000. Current Ratio = $₹12,00,000 \div ₹5,00,000 = 2.4:1$. Quick Assets = $₹12,00,000 - ₹5,00,000 - ₹50,000 = ₹6,50,000$; Quick Ratio = 1.3:1. Absolute Liquid Assets = $₹1,00,000 + ₹1,50,000 = ₹2,50,000$; Absolute Liquid Ratio = 0.5:1.

Question 6 — Profitability Ratios

Opening stock ₹2,00,000; net purchases ₹8,00,000; direct expenses ₹1,00,000; closing stock ₹3,00,000; sales ₹12,00,000. Calculate COGS, gross profit and gross profit ratio.

Solution: COGS = Opening Stock + Net Purchases + Direct Expenses – Closing Stock = $₹2,00,000 + ₹8,00,000 + ₹1,00,000 - ₹3,00,000 = ₹8,00,000$. Gross Profit = $₹12,00,000 - ₹8,00,000 = ₹4,00,000$. Gross Profit Ratio = $₹4,00,000 \div ₹12,00,000 \times 100 = 33.33\%$.

Question 7 — Profitability Ratios

Sales are ₹50,00,000 and EBIT is ₹7,50,000. Calculate the operating profit ratio. What does the result indicate?

Solution: Operating Profit Ratio = $EBIT \div Sales \times 100 = ₹7,50,000 \div ₹50,00,000 \times 100 = 15\%$. The business generates ₹15 of operating profit for every ₹100 of sales before interest and tax.

Question 8 — Profitability Ratios

Net sales are ₹40,00,000 and profit after tax is ₹3,20,000. Calculate the net profit ratio.

Solution: Net Profit Ratio = $Profit\ After\ Tax \div Net\ Sales \times 100 = ₹3,20,000 \div ₹40,00,000 \times 100 = 8\%$.

Question 9 — Profitability Ratios

Profit after interest and tax is ₹12,00,000. Preference dividend is ₹2,00,000 and the company has 5,00,000 equity shares. Calculate EPS.

Solution: EPS = $(Net\ Profit\ after\ Interest\ and\ Tax - Preference\ Dividend) \div Number\ of\ Equity\ Shares = (₹12,00,000 - ₹2,00,000) \div 5,00,000 = ₹2$ per equity share.

Question 10 — Profitability Ratios

Net profit after tax is ₹18,00,000, preference dividend ₹3,00,000 and equity shareholders' funds ₹1,00,00,000. Calculate ROE.

Solution: ROE = $(Net\ Profit\ after\ Tax - Preference\ Dividend) \div Equity\ Shareholders'\ Funds \times 100 = (₹18,00,000 - ₹3,00,000) \div ₹1,00,00,000 \times 100 = 15\%$.

Question 11 — Profitability Ratios

EBIT is ₹24,00,000. Debt is ₹80,00,000 and equity capital is ₹1,20,00,000. Calculate capital employed and ROCE.

Solution: Capital Employed = Debt + Equity = ₹80,00,000 + ₹1,20,00,000 = ₹2,00,00,000. ROCE = EBIT ÷ Capital Employed × 100 = ₹24,00,000 ÷ ₹2,00,00,000 × 100 = 12%.

Question 12 — Solvency Ratios

A company has long-term debt of ₹60,00,000 and equity of ₹40,00,000. Calculate the debt-equity ratio and debt-capital ratio.

Solution: Debt-Equity Ratio = Debt ÷ Equity = ₹60,00,000 ÷ ₹40,00,000 = 1.5:1. Debt-Capital Ratio = Debt ÷ (Debt + Equity) = ₹60,00,000 ÷ ₹1,00,00,000 = 0.60, or 60%.

Question 13 — Solvency Ratios

EBIT is ₹15,00,000 and annual interest expense is ₹3,00,000. Calculate the interest coverage ratio. Interpret the answer.

Solution: Interest Coverage Ratio = EBIT ÷ Interest = ₹15,00,000 ÷ ₹3,00,000 = 5 times. Operating earnings cover interest obligations five times, suggesting a relatively strong interest-paying capacity.

Question 14 — Turnover Ratios

Net credit sales are ₹72,00,000. Opening debtors are ₹5,00,000 and closing debtors ₹7,00,000. Calculate debtors turnover ratio and DSO.

Solution: Average Debtors = (₹5,00,000 + ₹7,00,000) ÷ 2 = ₹6,00,000. Debtors Turnover Ratio = ₹72,00,000 ÷ ₹6,00,000 = 12 times. DSO = 365 ÷ 12 = 30.42 days, approximately 30 days.

Question 15 — Turnover Ratios

Net credit purchases are ₹48,00,000. Opening payables are ₹3,00,000 and closing payables ₹5,00,000. Calculate creditors turnover ratio and DPO.

Solution: Average Payables = (₹3,00,000 + ₹5,00,000) ÷ 2 = ₹4,00,000. Creditors Turnover Ratio = ₹48,00,000 ÷ ₹4,00,000 = 12 times. DPO = 365 ÷ 12 = 30.42 days, approximately 30 days.

Question 16 — Turnover Ratios

COGS is ₹36,00,000. Opening inventory is ₹4,00,000 and closing inventory ₹5,00,000. Calculate inventory turnover ratio and DIO.

Solution: Average Inventory = (₹4,00,000 + ₹5,00,000) ÷ 2 = ₹4,50,000. Inventory Turnover Ratio = ₹36,00,000 ÷ ₹4,50,000 = 8 times. DIO = 365 ÷ 8 = 45.63 days, approximately 46 days.

Question 17 — Turnover Ratios

Revenue from operations is ₹1,20,00,000 and net fixed assets are ₹30,00,000. Calculate the fixed asset turnover ratio and explain it.

Solution: Fixed Asset Turnover Ratio = Revenue from Operations ÷ Net Fixed Assets = ₹1,20,00,000 ÷ ₹30,00,000 = 4 times. Each ₹1 invested in net fixed assets generates ₹4 of operating revenue.

Question 18 — Valuation Ratios

The market price of a share is ₹240 and EPS is ₹12. Calculate the P/E ratio. What does it mean?

Solution: P/E Ratio = Market Price per Share ÷ EPS = ₹240 ÷ ₹12 = 20 times. Investors are paying ₹20 for every ₹1 of current earnings per share.

Question 19 — Integrated Analysis

Company A has a current ratio of 2.2:1 and a quick ratio of 0.8:1. Company B has a current ratio of 1.6:1 and a quick ratio of 1.2:1. Which company appears more dependent on inventory for liquidity? Explain without additional calculation.

Solution: Company A is more dependent on inventory. Its current ratio is high, but the quick ratio falls sharply to 0.8:1 when inventory and prepaid expenses are excluded. Company B shows a smaller gap and therefore has a larger proportion of quick assets within current assets.

Question 20 — Integrated Analysis

A firm's DIO is 60 days, DSO is 45 days and DPO is 30 days. Calculate the cash conversion cycle (CCC). If DIO falls to 45 days with DSO and DPO unchanged, what happens to the CCC?

Solution: $CCC = DIO + DSO - DPO = 60 + 45 - 30 = 75$ days. After DIO falls to 45 days, $CCC = 45 + 45 - 30 = 60$ days. The CCC improves by 15 days, meaning cash is tied up in the operating cycle for a shorter period.

FORMULA RECAP

Current Ratio = Current Assets ÷ Current Liabilities

Quick Ratio = (Current Assets – Inventory – Prepaid Expenses) ÷ Current Liabilities

Absolute Liquid Ratio = (Cash + Bank + Short-term Investments) ÷ Current Liabilities

Gross Profit Ratio = Gross Profit ÷ Net Sales × 100

Operating Profit Ratio = EBIT ÷ Sales × 100

Net Profit Ratio = Profit After Tax ÷ Net Sales × 100

EPS = (Net Profit after Interest and Tax – Preference Dividend) ÷ Number of Equity Shares

ROE = (Net Profit after Tax – Preference Dividend) ÷ Equity Shareholders' Funds × 100

ROCE = EBIT ÷ Capital Employed × 100

Debt-Equity Ratio = Debt ÷ Equity

Debt-Capital Ratio = Debt ÷ (Debt + Equity)

Interest Coverage Ratio = EBIT ÷ Interest

Debtors Turnover = Net Credit Sales ÷ Average Debtors; DSO = 365 ÷ Debtors Turnover

Creditors Turnover = Net Credit Purchases ÷ Average Payables; DPO = 365 ÷ Creditors Turnover

Inventory Turnover = COGS ÷ Average Inventory; DIO = 365 ÷ Inventory Turnover

Fixed Asset Turnover = Revenue from Operations ÷ Net Fixed Assets

P/E Ratio = Market Price per Share ÷ EPS

Standard_Costing_Variance_Questions_with_Solutions.docx

Standard Costing – Variance Analysis

Practice Questions with Step-by-Step Solutions

Based on the uploaded BITS Pilani Financial and Management Accounting / Managerial Accounting presentation. The questions focus on Material Cost Variance, Material Price Variance, Material Usage Variance, Labour Cost Variance, Labour Rate Variance, and Labour Efficiency Variance. The presentation defines standard cost as a predetermined cost used to calculate variances and identify reasons for deviations.

Formula Sheet

Material Cost Variance (MCV): $MCV = \text{Standard Cost of Material} - \text{Actual Cost of Material} = (SQ \times SP) - (AQ \times AP)$

Material Price Variance (MPV): $MPV = AQ \times (SP - AP)$

Material Usage Variance (MUV): $MUV = SP \times (SQ - AQ)$

Check: $MPV + MUV = MCV$

Labour Cost Variance (LCV): $LCV = \text{Standard Cost of Labour} - \text{Actual Cost of Labour} = (ST \times SR) - (AT \times AR)$

Labour Rate Variance (LRV): $LRV = AT \times (SR - AR)$

Labour Efficiency Variance (LEV): $LEV = (ST - AT) \times SR$

Check: $LRV + LEV = LCV$

Note: SQ and ST are the standard quantity/time for the actual output, as emphasized in the presentation.

Section A – Material Variances

Question 1

A product has a standard material requirement of 5 kg per unit at a standard price of ₹20 per kg. Actual output is 200 units. Actual material used is 1,100 kg at ₹22 per kg. Calculate the Material Cost Variance.

Solution

Step 1: Standard quantity for actual output: $SQ = 5 \times 200 = 1,000 \text{ kg}$

Step 2: Standard cost: $SQ \times SP = 1,000 \times ₹20 = ₹20,000$

Step 3: Actual cost: $AQ \times AP = 1,100 \times ₹22 = ₹24,200$

Step 4: Material Cost Variance: $MCV = ₹20,000 - ₹24,200 = ₹4,200 \text{ Adverse}$

Answer: Material Cost Variance = ₹4,200 Adverse

Question 2

The standard material requirement is 4 kg per unit at ₹15 per kg. Actual production is 500 units. The company actually uses 1,900 kg at ₹16 per kg. Calculate MPV, MUV and MCV. Verify that $MPV + MUV = MCV$.

Solution

Step 1: Standard quantity: $SQ = 4 \times 500 = 2,000 \text{ kg}$

Step 2: MPV: $AQ \times (SP - AP) = 1,900 \times (₹15 - ₹16) = ₹1,900 \text{ Adverse}$

Step 3: MUV: $SP \times (SQ - AQ) = ₹15 \times (2,000 - 1,900) = ₹1,500$ Favourable

Step 4: MCV: $MPV + MUV = ₹1,900 \text{ A} + ₹1,500 \text{ F} = ₹400$ Adverse

Check using total costs: Standard cost = $2,000 \times ₹15 = ₹30,000$; Actual cost = $1,900 \times ₹16 = ₹30,400$.
MCV = ₹400 Adverse.

Answer: MPV = ₹1,900 A; MUV = ₹1,500 F; MCV = ₹400 A.

Question 3

A manufacturer sets a standard of 3 kg of material per unit at ₹12 per kg. Actual output is 800 units. Actual quantity used is 2,600 kg and actual price is ₹11 per kg. Calculate MPV, MUV and MCV.

Solution

Step 1: Standard quantity: $SQ = 3 \times 800 = 2,400$ kg

Step 2: MPV: $2,600 \times (₹12 - ₹11) = ₹2,600$ Favourable

Step 3: MUV: $₹12 \times (2,400 - 2,600) = ₹2,400$ Adverse

Step 4: MCV: $₹2,600 \text{ F} - ₹2,400 \text{ A} = ₹200$ Favourable

Answer: MPV = ₹2,600 F; MUV = ₹2,400 A; MCV = ₹200 F.

Question 4

The standard quantity of material for actual output is 1,500 kg and the standard price is ₹30 per kg. Actual quantity used is 1,600 kg at ₹28 per kg. Calculate the Material Cost Variance and split it into Price and Usage Variances.

Solution

Step 1: MCV: $(1,500 \times ₹30) - (1,600 \times ₹28) = ₹45,000 - ₹44,800 = ₹200$ Favourable

Step 2: MPV: $1,600 \times (₹30 - ₹28) = ₹3,200$ Favourable

Step 3: MUV: $₹30 \times (1,500 - 1,600) = ₹3,000$ Adverse

Step 4: Check: $₹3,200 \text{ F} - ₹3,000 \text{ A} = ₹200 \text{ F}$, which agrees with MCV.

Answer: MCV = ₹200 F; MPV = ₹3,200 F; MUV = ₹3,000 A.

Question 5

A company produced 300 units. The standard material requirement is 6 kg per unit at ₹25 per kg. Actual material consumption is 1,850 kg at ₹24 per kg. Calculate the standard quantity, standard cost, actual cost and all three material variances.

Solution

Step 1: SQ: $6 \times 300 = 1,800$ kg

Step 2: Standard cost: $1,800 \times ₹25 = ₹45,000$

Step 3: Actual cost: $1,850 \times ₹24 = ₹44,400$

Step 4: MCV: $₹45,000 - ₹44,400 = ₹600$ Favourable

Step 5: MPV: $1,850 \times (\text{₹}25 - \text{₹}24) = \text{₹}1,850$ Favourable

Step 6: MUV: $\text{₹}25 \times (1,800 - 1,850) = \text{₹}1,250$ Adverse

Step 7: Check: $\text{₹}1,850 \text{ F} - \text{₹}1,250 \text{ A} = \text{₹}600 \text{ F}$.

Answer: MCV = ₹600 F; MPV = ₹1,850 F; MUV = ₹1,250 A.

Section B – Labour Variances

Question 6

A product has a standard labour time of 2 hours per unit at a standard rate of ₹150 per hour. Actual output is 400 units. Workers actually take 850 hours at ₹160 per hour. Calculate Labour Cost Variance, Labour Rate Variance and Labour Efficiency Variance.

Solution

Step 1: Standard time: $ST = 2 \times 400 = 800$ hours

Step 2: Standard labour cost: $800 \times \text{₹}150 = \text{₹}1,20,000$

Step 3: Actual labour cost: $850 \times \text{₹}160 = \text{₹}1,36,000$

Step 4: LCV: $\text{₹}1,20,000 - \text{₹}1,36,000 = \text{₹}16,000$ Adverse

Step 5: LRV: $850 \times (\text{₹}150 - \text{₹}160) = \text{₹}8,500$ Adverse

Step 6: LEV: $(800 - 850) \times \text{₹}150 = \text{₹}7,500$ Adverse

Step 7: Check: $\text{₹}8,500 \text{ A} + \text{₹}7,500 \text{ A} = \text{₹}16,000 \text{ A}$.

Answer: LCV = ₹16,000 A; LRV = ₹8,500 A; LEV = ₹7,500 A.

Question 7

The standard labour time for actual production is 1,200 hours at ₹100 per hour. Actual labour time is 1,100 hours and workers are paid ₹105 per hour. Calculate LCV, LRV and LEV.

Solution

Step 1: LCV: $(1,200 \times \text{₹}100) - (1,100 \times \text{₹}105) = \text{₹}1,20,000 - \text{₹}1,15,500 = \text{₹}4,500$ Favourable

Step 2: LRV: $1,100 \times (\text{₹}100 - \text{₹}105) = \text{₹}5,500$ Adverse

Step 3: LEV: $(1,200 - 1,100) \times \text{₹}100 = \text{₹}10,000$ Favourable

Step 4: Check: $\text{₹}10,000 \text{ F} - \text{₹}5,500 \text{ A} = \text{₹}4,500 \text{ F}$.

Answer: LCV = ₹4,500 F; LRV = ₹5,500 A; LEV = ₹10,000 F.

Question 8

A factory sets a standard of 5 labour hours per unit at ₹80 per hour. Actual output is 250 units. Actual hours worked are 1,300 and the actual labour rate is ₹75 per hour. Calculate the labour variances.

Solution

Step 1: Standard time: $ST = 5 \times 250 = 1,250$ hours

Step 2: LCV: $(1,250 \times ₹80) - (1,300 \times ₹75) = ₹1,00,000 - ₹97,500 = ₹2,500$ Favourable

Step 3: LRV: $1,300 \times (₹80 - ₹75) = ₹6,500$ Favourable

Step 4: LEV: $(1,250 - 1,300) \times ₹80 = ₹4,000$ Adverse

Step 5: Check: $₹6,500 \text{ F} - ₹4,000 \text{ A} = ₹2,500 \text{ F}$.

Answer: LCV = ₹2,500 F; LRV = ₹6,500 F; LEV = ₹4,000 A.

Question 9

The standard labour cost for actual production is ₹90,000. The actual labour cost is ₹96,000. The standard time for actual production is 1,000 hours, while actual time is 1,200 hours. Calculate the standard labour rate, actual labour rate and then calculate LRV and LEV.

Solution

Step 1: Standard rate: $SR = ₹90,000 \div 1,000 = ₹90$ per hour

Step 2: Actual rate: $AR = ₹96,000 \div 1,200 = ₹80$ per hour

Step 3: LCV: $₹90,000 - ₹96,000 = ₹6,000$ Adverse

Step 4: LRV: $1,200 \times (₹90 - ₹80) = ₹12,000$ Favourable

Step 5: LEV: $(1,000 - 1,200) \times ₹90 = ₹18,000$ Adverse

Step 6: Check: $₹12,000 \text{ F} - ₹18,000 \text{ A} = ₹6,000 \text{ A}$.

Answer: LCV = ₹6,000 A; LRV = ₹12,000 F; LEV = ₹18,000 A.

Question 10

A company has a standard labour time of 4 hours per unit at ₹125 per hour. Actual production is 150 units. Workers take 570 hours and are paid ₹120 per hour. Calculate LCV, LRV and LEV. Interpret the result.

Solution

Step 1: Standard time: $ST = 4 \times 150 = 600$ hours

Step 2: Standard cost: $600 \times ₹125 = ₹75,000$

Step 3: Actual cost: $570 \times ₹120 = ₹68,400$

Step 4: LCV: $₹75,000 - ₹68,400 = ₹6,600$ Favourable

Step 5: LRV: $570 \times (₹125 - ₹120) = ₹2,850$ Favourable

Step 6: LEV: $(600 - 570) \times ₹125 = ₹3,750$ Favourable

Step 7: Check: $₹2,850 \text{ F} + ₹3,750 \text{ F} = ₹6,600 \text{ F}$.

Interpretation: The favourable total labour variance results from both a lower actual labour rate and fewer hours used than the standard.

Answer: LCV = ₹6,600 F; LRV = ₹2,850 F; LEV = ₹3,750 F.

Section C – Integrated Questions

Question 11

A company manufactures 1,000 units. Standard material requirement is 2 kg per unit at ₹30 per kg. Actual material used is 2,100 kg at ₹29 per kg. Standard labour time is 3 hours per unit at ₹100 per hour. Actual labour time is 2,900 hours at ₹105 per hour. Calculate all material and labour variances.

Solution

Material SQ: $2 \times 1,000 = 2,000$ kg

Material MPV: $2,100 \times (\text{₹}30 - \text{₹}29) = \text{₹}2,100$ F

Material MUV: $\text{₹}30 \times (2,000 - 2,100) = \text{₹}3,000$ A

Material MCV: $\text{₹}2,100$ F – $\text{₹}3,000$ A = $\text{₹}900$ A

Labour ST: $3 \times 1,000 = 3,000$ hours

Labour LRV: $2,900 \times (\text{₹}100 - \text{₹}105) = \text{₹}14,500$ A

Labour LEV: $(3,000 - 2,900) \times \text{₹}100 = \text{₹}10,000$ F

Labour LCV: $\text{₹}10,000$ F – $\text{₹}14,500$ A = $\text{₹}4,500$ A

Answer: Material: MCV ₹900 A, MPV ₹2,100 F, MUV ₹3,000 A. Labour: LCV ₹4,500 A, LRV ₹14,500 A, LEV ₹10,000 F.

Question 12

For a particular month, the standard material cost for actual output is ₹60,000 and actual material cost is ₹63,500. The material price variance is ₹2,500 Adverse. Determine the material usage variance.

Solution

Step 1: Relationship: $\text{MPV} + \text{MUV} = \text{MCV}$

Step 2: MCV: $\text{₹}60,000 - \text{₹}63,500 = \text{₹}3,500$ Adverse

Step 3: Solve for MUV: $\text{MUV} = \text{MCV} - \text{MPV} = \text{₹}3,500 \text{ A} - \text{₹}2,500 \text{ A} = \text{₹}1,000 \text{ A}$

Answer: Material Usage Variance = ₹1,000 Adverse.

Question 13

For a production period, the standard labour cost for actual production is ₹50,000 and the actual labour cost is ₹47,000. The labour efficiency variance is ₹5,000 Favourable. Calculate the labour rate variance.

Solution

Step 1: LCV: $\text{₹}50,000 - \text{₹}47,000 = \text{₹}3,000$ Favourable

Step 2: Relationship: $\text{LRV} + \text{LEV} = \text{LCV}$

Step 3: Solve for LRV: $\text{LRV} = \text{₹}3,000 \text{ F} - \text{₹}5,000 \text{ F} = \text{₹}2,000$ Adverse

Answer: Labour Rate Variance = ₹2,000 Adverse.

Key Reminders from the class

Standard quantity (SQ) must be based on actual output, not budgeted output.

Standard time (ST) must be based on actual production.

For material variances: $MPV + MUV = MCV$.

For labour variances: $LRV + LEV = LCV$.

A positive variance in the presentation's formulation (standard cost minus actual cost) represents a favourable result; a negative result represents an adverse result.

PV ratio practice question.docx

A company sells a product at ₹600 per unit. Variable cost is ₹360 per unit. Fixed costs are ₹4,80,000. Calculate: (a) Break-even point in units (b) Break-even sales value

Solution:

$SP = \text{Rs } 600$ Contribution/unit = $SP - VC = 600 - 360 = ₹240$ BEP (units) = $FC / \text{Contribution margin per unit}$
 $4,80,000 / 240 = 2,000$ units BEP (sales) = $SP * \text{No. of units for breakeven} = 2,000 \times 600 = ₹12,00,000$

Selling price = ₹500, variable cost = ₹320, fixed cost = ₹6,30,000. Find units required to earn profit of ₹2,70,000.

$BEP = FC / \text{Contribution per unit}$

No of units required to earn Rs. 270000 = $(630000 + 270000) / 180 = 5000$

$BEP = \text{Total FC} / \text{Contribution margin per unit}$

$630000 / 180 = 3500$

$(630000 + 270000) / 180 = 5000$

Total Revenue = $5000 * 500 = 250000$

$P/V \text{ Ratio} = (\text{Sales} - VC) / \text{Sales}$ or $(SP - VC \text{ per unit}) / SP$

Solution: Contribution/unit = $500 - 320 = ₹180$ Required units = $(6,30,000 + 2,70,000) / 180 = 5,000$ units

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Sales = ₹25,00,000, variable cost = ₹15,00,000, fixed cost = ₹6,00,000. Find P/V ratio and profit.

Solution: Contribution = $10,00,000 = 25 \text{ lakhs} - 15 \text{ lakhs}$ P/V ratio = $10,00,000 / 25,00,000 = 40\%$ Contri - FC = Profit

Profit = $10,00,000 - 6,00,000 = ₹4,00,000$

Break-even sales are ₹18,00,000 and P/V ratio is 35%. Find fixed cost.

$PV/\text{Ratio} = FC / \text{Break Even Sales}$

$FC = PV/\text{Ratio} * \text{Sales}$

Solution: Fixed cost = $18,00,000 \times 35\% = ₹6,30,000$

Sales = ₹40,00,000, variable costs = ₹24,00,000, fixed costs = ₹10,00,000. Calculate margin of safety.
BEP (Sales)

Solution: Contribution = 16,00,000

Sales = 40,00,000 Profit = 6,00,000 P/V ratio = 40% BE Sales = 10,00,000 / 0.40 = ₹25,00,000 MOS = 40,00,000 - 25,00,000 = ₹15,00,000

A company produces two products A and B in sales mix 3:2. Contribution/unit A = ₹50, B = ₹80. Fixed cost ₹9,20,000. Find composite BEP.

Solution: Contribution of bundle = (3×50)+(2×80)=310 Bundles = 9,20,000 / 310 = 2,968 bundles

Selling price ₹800, variable cost ₹500. Fixed cost ₹12,00,000. If price reduced by 10%, find new BEP units.
Solution: New SP = 720 Contribution = 720 - 500 = 220 BEP = 12,00,000 / 220 = 5,455 units

Variable cost ₹300 rises by 15%. SP ₹500. Fixed cost ₹5,75,000. Find revised BEP units. Solution: New VC = 345 Contribution = 155 BEP = 5,75,000 / 155 = 3,710 units

Sales ₹50,00,000, profit ₹8,00,000, margin of safety ₹10,00,000. Find BE sales and P/V ratio. Solution: BE Sales = 50,00,000 - 10,00,000 = ₹40,00,000 Profit = MOS × P/V ratio 8,00,000 = 10,00,000 × ratio Ratio = 80%

Fixed cost rises from ₹4,00,000 to ₹5,50,000. Contribution/unit = ₹75. Find increase in BEP units.
Solution:

Old BEP = 4,00,000/75 = 5,333 New BEP = 5,50,000/75 = 7,333 Increase = 2,000 units

Sales ₹30,00,000, variable cost ratio 60%, fixed cost ₹7,50,000. Find profit and operating leverage.
Solution: Contribution = 12,00,000 Profit = 4,50,000

Contribution/Profit = Operating Leverage (OL) OL = 12,00,000 / 4,50,000 = 2.67

A firm earns ₹6,00,000 profit at sales ₹36,00,000. P/V ratio = 25%. Find fixed cost. Solution:

P/V Ratio= Contribution/ Sales = 25% = Contri/ 36 36*25% = 9 lakhs Profit = Sales - VC- FC = Contribution - FC

Contribution = 9,00,000 Fixed cost = 9,00,000 - 6,00,000 = ₹3,00,000

If contribution margin ratio is 30% and fixed costs ₹9,00,000, calculate sales for profit of ₹6,00,000. Profit = Contribution - FC = Profit + FC = Contribution = 9+6 = 15 lakhs

Contribution Margin Ratio = Total Contribution/ Total Sales *100 15,00,000 / Sales = 30% = 15 lakhs/30% = 50 lakhs

Solution: Required sales = (9,00,000+6,00,000)/0.30 = ₹50,00,000

Product sells ₹1,000, VC ₹650. Fixed cost ₹14,00,000. Find sales units for after-tax profit ₹4,20,000 (tax 30%). Let before tax profit = 100 Tax = 30 After tax profit = 70

Solution: Pre-tax profit = 4,20,000 / 0.70 = 6,00,000

Profit = Contribution - FC = 6,00,000+14,00,000 = 20,00,000 Required contribution = 20,00,000 Units = 20,00,000 / 350 = 5,715 units

A company has sales ₹60,00,000 and contribution ₹18,00,000. If fixed cost ₹12,00,000, calculate MOS.

Solution: PV/Ratio = Contribution / Sales

Profit = 6,00,000 P/V ratio = 30% MOS = Profit / P/V ratio = 6,00,000/0.30 = ₹20,00,000

Machine A fixed cost ₹5,00,000 contribution/unit ₹40. Machine B fixed cost ₹8,00,000 contribution/unit ₹60. Find indifference point.

Solution: Contribution = Sales - VC - FC

$$40Q - 5,00,000 = 60Q - 8,00,000 \quad 3,00,000 = 20Q \quad Q = 15,000 \text{ units}$$

Sales mix X:Y = 1:3. SP X=₹200, VC=₹120. SP Y=₹300, VC=₹210. Fixed cost ₹7,20,000. Find BE bundles.

Solution: Contribution X=80, Y=90 Bundle contribution = $1 \times 80 + 3 \times 90 = 350$ Bundles = $7,20,000 / 350 = 2,057$ bundles

Present sales 20,000 units. SP ₹150, VC ₹90, FC ₹7,00,000. If sales rise 20%, estimate profit. Solution: Contribution/unit = 60 Current profit = 12,00,000 - 7,00,000 = 5,00,000 New units = 24,000 New profit = 14,40,000 - 7,00,000 = ₹7,40,000

A company has degree of operating leverage 4 and profit ₹3,00,000. Find contribution and fixed cost. Solution:

$$OL = \text{Contri/Profit} = \text{Contri}/(\text{Contri}-\text{FC}) \quad \text{Contribution} = 4 \times 3,00,000 = ₹12,00,000 \quad \text{Fixed cost} = 12,00,000 - 3,00,000 = ₹9,00,000$$

Sales decline by 15%. Current sales ₹80,00,000, P/V ratio 35%, fixed cost ₹20,00,000. Find revised profit. Solution: New sales = 68,00,000

$$\text{Contri/New Sales} = \text{PV ratio} \quad \text{Contribution} = 23,80,000 \quad \text{Profit} = 23,80,000 - 20,00,000 = ₹3,80,000$$

Advanced_Financial_Ratio_Analysis_20_Classroom_Problems_Solutions.docx

FINANCIAL RATIO ANALYSIS 20 ADVANCED CLASSROOM DISCUSSION PROBLEMS WITH SOLUTIONS

Application • Interpretation • Comparison • Decision-Making

Classroom approach: Ask students to calculate first, interpret the result, and then defend a managerial or investor conclusion. Use 365 days unless otherwise stated.

Question 1: Liquidity under pressure

Zenith Ltd. reports cash ₹1.2 lakh, bank ₹0.8 lakh, marketable securities ₹1 lakh, debtors ₹5 lakh, inventory ₹7 lakh and prepaid expenses ₹1 lakh. Current liabilities are ₹8 lakh. Calculate the current, quick and absolute liquid ratios. The CFO claims, "Our current ratio proves that liquidity is strong." Evaluate the claim.

Suggested solution and discussion: Current assets = ₹16 lakh. Current ratio = $16/8 = 2:1$. Quick assets = $16 - 7 - 1 = ₹8$ lakh; quick ratio = 1:1. Absolute liquid assets = $1.2 + 0.8 + 1 = ₹3$ lakh; absolute liquid ratio = $3/8 = 0.375:1$. The CFO's conclusion is incomplete. The current ratio looks comfortable, but a large part of current assets is inventory. Immediate cash coverage is only 0.375 times current liabilities. Students should distinguish accounting liquidity from immediate cash availability.

Question 2: Window dressing the current ratio

A firm has current assets of ₹20 lakh and current liabilities of ₹10 lakh. Just before year-end, it uses ₹4 lakh cash to pay creditors. Calculate the current ratio before and after the transaction. Has the underlying business performance improved?

Suggested solution and discussion: Before payment, current ratio = $20/10 = 2:1$. After payment, current assets = ₹16 lakh and current liabilities = ₹6 lakh. New current ratio = $16/6 = 2.67:1$. The ratio improves mechanically, but no sales, profit or operating efficiency has improved. This is a useful discussion on how year-end transactions can alter ratios and why ratios require context.

Question 3: Inventory-heavy liquidity

Company A has current assets ₹30 lakh, inventory ₹18 lakh, prepaid expenses ₹2 lakh and current liabilities ₹15 lakh. Company B has current assets ₹24 lakh, inventory ₹5 lakh, prepaid expenses ₹1 lakh and current liabilities ₹15 lakh. Compare liquidity using current and quick ratios. Which firm is financially safer in the short run?

Suggested solution and discussion: A: Current ratio = $30/15 = 2:1$; quick ratio = $(30-18-2)/15 = 0.67:1$. B: Current ratio = $24/15 = 1.6:1$; quick ratio = $(24-5-1)/15 = 1.2:1$. Although A has the better current ratio, B has stronger quick liquidity. A is highly dependent on selling inventory. The safer firm depends partly on inventory quality and turnover, but B appears stronger on immediate liquidity.

Question 4: Gross margin deterioration

A company's sales rise from ₹100 lakh to ₹130 lakh. Gross profit rises from ₹35 lakh to ₹39 lakh. Calculate the gross profit ratio for both years. The sales director calls the year a major success. Discuss.

Suggested solution and discussion: Year 1 GP ratio = $35/100 \times 100 = 35\%$. Year 2 = $39/130 \times 100 = 30\%$. Sales grew 30%, and absolute gross profit increased, but margin fell by 5 percentage points. Possible reasons include discounting, input-cost inflation or a less profitable product mix. The sales director's claim should be assessed alongside margin quality.

Question 5: From gross profit to EBIT

Nova Ltd. has sales ₹200 lakh. Opening inventory ₹20 lakh, net purchases ₹105 lakh, direct expenses ₹15 lakh and closing inventory ₹25 lakh. Administrative and selling expenses are ₹45 lakh. Calculate COGS, gross profit, gross profit ratio, EBIT and operating profit ratio.

Suggested solution and discussion: COGS = $20 + 105 + 15 - 25 = ₹115$ lakh. Gross profit = $200 - 115 = ₹85$ lakh. GP ratio = 42.5% . EBIT = $85 - 45 = ₹40$ lakh. Operating profit ratio = $40/200 \times 100 = 20\%$. The problem helps students connect the trading result with operating profitability.

Question 6: Same sales, different profitability

Company X and Company Y each report sales of ₹500 lakh. X has gross profit ₹200 lakh and EBIT ₹50 lakh. Y has gross profit ₹150 lakh and EBIT ₹75 lakh. Calculate GP and operating profit ratios. Which company performs better?

Suggested solution and discussion: X: GP ratio = 40% ; operating profit ratio = 10% . Y: GP ratio = 30% ; operating profit ratio = 15% . X earns a higher product/trading margin but loses more through operating expenses. Y converts sales into operating profit more efficiently. There is no single answer to 'better'; students should identify the operational issue in each company.

Question 7: EPS and preference shareholders

A company earns profit after interest and tax of ₹30 lakh. It has ₹5 lakh preference dividend and 10 lakh equity shares. The next year, profit rises to ₹36 lakh, preference dividend remains unchanged and the company issues 5 lakh additional equity shares. Calculate EPS for both years. Did equity shareholders necessarily benefit from higher profit?

Suggested solution and discussion: Year 1 EPS = $(30-5)/10 = ₹2.50$. Year 2 EPS = $(36-5)/15 = ₹2.07$ approximately. Profit increased by 20%, but EPS declined because the number of equity shares increased substantially. Higher total profit does not automatically mean higher earnings per equity share.

Question 8: ROE versus ROCE

Alpha Ltd. has EBIT ₹24 lakh, interest ₹6 lakh and tax ₹4 lakh. Debt is ₹60 lakh and equity shareholders' funds ₹60 lakh. Assume no preference dividend. Calculate ROCE and ROE. Explain why they differ.

Suggested solution and discussion: Capital employed = 60 + 60 = ₹120 lakh. $ROCE = 24/120 \times 100 = 20\%$. Profit after tax = 24 – 6 – 4 = ₹14 lakh. $ROE = 14/60 \times 100 = 23.33\%$. ROCE measures returns generated by total long-term capital before interest and tax, while ROE focuses on returns available to equity holders after financing and tax effects.

Question 9: Leverage comparison

Firm P has debt ₹20 crore and equity ₹80 crore. Firm Q has debt ₹70 crore and equity ₹30 crore. Both have EBIT ₹15 crore. Calculate debt-equity and debt-capital ratios. Which firm is more financially risky? What additional ratio would you request?

Suggested solution and discussion: P: $D/E = 20/80 = 0.25:1$; debt-capital = $20/100 = 20\%$. Q: $D/E = 70/30 = 2.33:1$; debt-capital = $70/100 = 70\%$. Q is more leveraged and potentially more financially risky. However, students should request the interest coverage ratio because the cost of debt and ability to service interest are essential.

Question 10: Interest coverage shock

A business has EBIT ₹40 lakh and interest ₹5 lakh. Due to a downturn, EBIT falls by 50%, while interest rises to ₹8 lakh after refinancing. Calculate interest coverage before and after. Discuss the risk.

Suggested solution and discussion: Initial coverage = $40/5 = 8$ times. Revised EBIT = ₹20 lakh; revised coverage = $20/8 = 2.5$ times. Coverage falls sharply from 8 to 2.5 times. The company has a much smaller operating cushion for interest payments and is more vulnerable to another earnings decline.

Question 11: Receivables warning

Credit sales increase from ₹120 lakh to ₹180 lakh. Average debtors increase from ₹15 lakh to ₹30 lakh. Calculate debtors turnover and DSO for both years. Has collection efficiency kept pace with sales growth?

Suggested solution and discussion: Year 1 turnover = $120/15 = 8$ times; $DSO = 365/8 = 45.63$ days. Year 2 turnover = $180/30 = 6$ times; $DSO = 365/6 = 60.83$ days. Sales increased 50%, but debtors doubled. Collection slowed by about 15 days. Growth may be partly supported by looser credit terms or weaker collection.

Question 12: Credit policy decision

A company currently has credit sales ₹365 lakh and average debtors ₹45 lakh. Management proposes relaxed credit terms expected to increase sales to ₹438 lakh, but average debtors may rise to ₹73 lakh. Calculate DSO before and after. Identify one benefit and one risk of the proposal.

Suggested solution and discussion: Current turnover = $365/45 = 8.11$ times; $DSO \approx 45$ days. Proposed turnover = $438/73 = 6$ times; $DSO \approx 60.83$ days. Benefit: higher sales. Risk: cash is locked in receivables longer and bad-debt or financing costs may rise. Students should not judge the policy on sales growth alone.

Question 13: Supplier financing

Net credit purchases are ₹240 lakh. Average payables rise from ₹20 lakh to ₹40 lakh while purchases remain unchanged. Calculate creditors turnover and DPO in both situations. Is the increase in DPO automatically positive?

Suggested solution and discussion: Initially, turnover = $240/20 = 12$ times; DPO = 30.42 days. Later, turnover = $240/40 = 6$ times; DPO = 60.83 days. Longer DPO conserves cash and provides supplier financing, but it may also signal payment difficulty or damage supplier relationships. Therefore, it is not automatically positive.

Question 14: Inventory efficiency or stock-out risk?

COGS is ₹360 lakh. Average inventory falls from ₹60 lakh to ₹30 lakh. Calculate inventory turnover and DIO before and after. Management celebrates the improvement. What risk should students investigate?

Suggested solution and discussion: Initially, inventory turnover = $360/60 = 6$ times; DIO = 60.83 days. After reduction, turnover = $360/30 = 12$ times; DIO = 30.42 days. Efficiency appears to improve substantially. However, students should investigate stock-outs, lost sales, production interruptions and whether inventory reduction is sustainable.

Question 15: Working-capital cycle

A retailer has DIO 75 days, DSO 35 days and DPO 50 days. Calculate the cash conversion cycle. Management can either reduce DIO by 15 days or increase DPO by 15 days. Show the effect of each option and discuss whether the financial effect is identical in practice.

Suggested solution and discussion: CCC = $75 + 35 - 50 = 60$ days. Reducing DIO by 15 days gives CCC = 45 days. Increasing DPO by 15 days also gives CCC = 45 days. Mathematically the effect is identical, but operational consequences differ: lower inventory may cause stock-outs, while delayed supplier payment may affect relationships or discounts.

Question 16: Fixed asset productivity

Company M and N each generate revenue of ₹300 crore. M has net fixed assets ₹75 crore and N ₹150 crore. Calculate fixed asset turnover. Can you conclude that M is the better company?

Suggested solution and discussion: M turnover = $300/75 = 4$ times. N = $300/150 = 2$ times. M generates more revenue per rupee of fixed assets. However, M may operate an asset-light model, use leased assets, or have older depreciated assets. N may have recently invested in capacity. Ratio interpretation requires business-model and asset-age context.

Question 17: P/E puzzle

Stock A trades at ₹300 with EPS ₹15. Stock B trades at ₹240 with EPS ₹8. Calculate the P/E ratios. Which stock is cheaper based only on P/E? Why is a lower P/E not automatically a better investment?

Suggested solution and discussion: A P/E = $300/15 = 20$ times. B P/E = $240/8 = 30$ times. Based only on P/E, A is cheaper relative to current earnings. However, B's higher P/E may reflect stronger expected growth, lower risk or superior business quality. A low P/E may also reflect weak future prospects.

Question 18: Ratio chain diagnosis

A company reports the following changes: current ratio 2.1 to 2.4, quick ratio 1.3 to 0.9, inventory turnover 8 to 5 times and gross profit ratio 32% to 36%. Diagnose what may be happening.

Suggested solution and discussion: The higher current ratio alone looks positive, but the falling quick ratio and inventory turnover suggest inventory accumulation. The higher gross margin may reflect price increases or a higher-margin product mix, but unsold inventory may also be building. Students should examine inventory ageing, demand, production levels and sales trends.

Question 19: Investor comparison

Company R: ROE 24%, debt-equity 2.5:1, interest coverage 2 times, P/E 12. Company S: ROE 16%, debt-equity 0.4:1, interest coverage 9 times, P/E 22. Without calculating further ratios, compare the two from the perspective of a conservative investor.

Suggested solution and discussion: R offers higher ROE and a lower P/E, but it carries much higher leverage and weak interest coverage. Its ROE may be amplified by debt. S has lower ROE and a higher valuation but much stronger solvency. A conservative investor may prefer S, subject to growth and industry analysis. The key learning is that high ROE should not be read in isolation.

Question 20: Comprehensive boardroom case

Orion Ltd. reports: sales ₹600 lakh; COGS ₹390 lakh; EBIT ₹72 lakh; PAT ₹36 lakh; current assets ₹180 lakh, including inventory ₹90 lakh and prepaid expenses ₹6 lakh; current liabilities ₹90 lakh; debt ₹180 lakh; equity ₹120 lakh; interest ₹24 lakh; average debtors ₹75 lakh; all sales are credit. Calculate the GP ratio, operating profit ratio, net profit ratio, current ratio, quick ratio, debt-equity ratio, interest coverage, debtors turnover and DSO. Give a three-point boardroom diagnosis.

Suggested solution and discussion: GP ratio = $(600 - 390) / 600 \times 100 = 35\%$. Operating profit ratio = $72 / 600 \times 100 = 12\%$. Net profit ratio = $36 / 600 \times 100 = 6\%$. Current ratio = $180 / 90 = 2:1$. Quick ratio = $(180 - 90 - 6) / 90 = 0.93:1$. Debt-equity = $180 / 120 = 1.5:1$. Interest coverage = $72 / 24 = 3$ times. Debtors turnover = $600 / 75 = 8$ times; DSO = 45.63 days. Diagnosis: (1) current liquidity looks adequate but quick liquidity is below 1 because inventory is significant; (2) leverage is meaningful and interest coverage of 3 times needs monitoring; (3) gross margin is 35%, but only 6% reaches PAT, so operating, financing and tax costs should be analysed.

FACULTY DISCUSSION PROMPTS

Which ratio gives the first warning signal? Why?

Can management improve this ratio mechanically without improving the business?

What additional information would you request before making a decision?

Would a lender, investor and operating manager interpret the same ratio differently?

Is the change driven by the numerator, denominator or both?

Could industry characteristics explain an apparently weak ratio?

What trade-off is management making between profitability, liquidity and risk?

Ribbon an' Bows Case Study.pdf

(no extractable text — likely a scanned/image PDF; open the original file directly)

Ribbons and bows-Solution.xlsx

BALANCE SHEET ON 31ST MARCH-GIVEN | | | | | | | CASH BALANCE ON 31ST MARCH

ASSETS | | LIABILITIES | | | | | LOAN | 10000

PREPAID RENT | 1200 | LOAN | 10000 | | | | EQUITY | 1000

INVENTORY | 3300 | EQUITY | 1000 | | | | |

SUPPLIES | 100 | | | | | | LESS: OUTFLOW |

CASH | 4000 | | | | | |

OREPAID ADVT | 150 | | | | | | PREPAID RENT | 1200

CASH DEPOSIT REGISTER | 250 | | | | | | INVENTORY | 3300

SOFTWARE | 2000 | | | | | | SUPPLIES | 100

| 11000 | | 11000 | | | | | SOFTWARE | 2000

| | | | | | | OREPAID ADVT | 150

| | | | | | | CASH DEPOSIT REGISTER | 250

ALTERNATIVELY | | | | | | CASH BALANCE ON 31ST MARCH | 4000

TRIAL BALANCE ON 31ST MARCH -GIVEN | | | | | | |

PARTICULARS | DR | CR | | | | | |

LOAN | | 10000 | | | | | |

EQUITY | | 1000 | | | | | |

PREPAID RENT | 1200 | | | | | | |

INVENTORY | 3300 | | | | | | |

SUPPLIES | 100 | | | | | | |

CASH | 4000 | | | | | | |

OREPAID ADVT | 150 | | | | | | |

CASH DEPOSIT REGISTER | 250 | | | | | | |

SOFTWARE | 2000 | | | | | | |

| 11000 | 11000 | | | | | |

INCOME STATEMENTFROM MAR1ST TO 30TH JUNE | | | | | | | BALANCE SHEET ON 31ST MARCH-GIVEN | |
|

| | | | | | | ASSETS | | LIABILITIES |

SALES | 7720 | | | | | | | PREPAID RENT | 1200 | LOAN | 10000

| | | | | | | INVENTORY | 3300 | EQUITY | 1000

COGS | 2100 | | | | | | | SUPPLIES | 100 | |

| | | | | | | CASH | 4000 | |

GP | 5620 | | | | | | | OREPAID ADVT | 150 | |

| | | | | | | CASH DEPOSIT REGISTER | 250 | |

RENT (3000-1200) | 1800 | | | | | | | SOFTWARE | 2000 | |

SALARY | 1600 | | | | | | | 11000 | | 11000

adv-t-(prepaid) | 150 | | | | | | | |

OFFICE SUPPLIES | 80 |||||

EBITDA | 1990 |||||

DEPN |||||

SM | 60 ||||| CASH BALANCE ON 31ST MARCH ||

COMPUTER | 250 ||||| LOAN | 10000 ||

||||| EQUITY | 1000 ||

OP PROFIT/EBIT | 1680 |||||

INT ON LOAN-FROM MAR 1ST | 200 ||||| LESS: OUTFLOW |||

EBT | 1480 |||||

LESS:TAX | 0 ||||| PREPAID RENT | 1200 ||

PAT | 1480 ||||| INVENTORY | 3300 ||

||||| SUPPLIES | 100 ||

||||| SOFTWARE | 2000 ||

||||| OREPAID ADVT | 150 ||

BALANCE SHEET ON 30TH JUNE ||||| CASH DEPOSIT REGISTER | 250 ||

ASSETS || LIABILITIES AND EQUITY |||||

SEWING MACHINE | 1740 | LOAN | 10000 ||||| CASH BALANCE ON 31ST MARCH | 4000 ||

DEBTORS | 320 | EQUITY | 1000 |||||

INVENTORY | 4100 | profit | 1480 |||||

OFFICE SUPPLIES BALANCE | 20 | O/S SALARY | 90 |||||

CASH | 3390 | O/S INT | 200 |||||

SOFTWARE | 1750 |||||

CASH DEPOSIT REGISTER | 250 |||||

PREPAID RENT | 1200 |||||

| 12770 || 12770 |||||

CASH BALANCE ON 30TH JUNE |||||

OP BAL | 4000 |||||

CASH RECD FROM SALES | 7400 |||||

| 11400 |||||

SM | 1800 |||||

SALARY PAID | 1510 |||||

RENT PAID | 1800 |||||

INVENTORY | 2900 |||||

| 3390 |||||

Arjun has always been passionate about organic farming and decides to start his own organic grocery store. He invests ₹200,000 from his savings and takes a loan of ₹50,000 from his friend. His friend charges an interest rate of 6%, which is lower than the market rate of 8%. The friend insists that the loan should be interest-free for the first one year and suggests that interest on the loan can be charged from the end of the first year. They document the loan agreement for clarity. | |

Arjun purchases shelves, a refrigerator, and initial stock of organic produce. He hires two assistants at ₹6,000 per month each, starting from 1st April 2024. He registers his business as Arjun Organics Pvt Ltd, where he and his sister are equity shareholders. | |

Trial Balance as on 1st April 2024: | |

Particulars | Dr. (₹) | Cr. (₹)

Shelves | 12000 |

Refrigerator | 70000 |

Initial Stock (Inventory) | 40000 |

Loan from Friend @6% | | 50000

Capital | | 200000

Cash in Hand | 128000 |

Total | 250000 | 250000

At the end of six months (September 30, 2024), Arjun wants to assess his financial standing. He asks his friend to help him prepare a Profit & Loss Account and Balance Sheet for the six months. | |

Transactions during April 2024 – September 2024: | |

- i. Arjun purchases fresh organic stock worth ₹30,000 on 30th May 2024, on credit and ₹40,000 on 30th July 2024, paying in cash. | |
- ii. At the end of six months, he still has stock worth ₹15,000. | |
- iii. He earns ₹150,000 from cash sales over the six months. | |
- iv. Arjun supplies organic produce to restaurants on credit. The total revenue from these sales is ₹220,000, but ₹50,000 remains outstanding as of September 30, 2024. | |
- v One of the assistants has been paid a salary advance for October 2024. | |
- vi He pays ₹3,000 per month as rent, totaling ₹18,000 for six months. | |
- vii. Annual depreciation charged on shelves is 5% pa. | |
- viii Arjun’s sister helps with the business on a regular basis. She has been into farming for a long time and her contacts have helped to boost sales. Arjun estimates that his sister’s contacts would have

helped boost sales by about 30%. He has not paid his sister a salary for her services and believes she should get a monthly salary that is at least double that of the assistants. | |

Requirements: | |

a) Prepare a Profit & Loss Account and Balance Sheet of Arjun Organics Pvt Ltd for the six months ending September 30, 2024. | |

b) Calculate the cash balance Arjun has in hand as of September 30, 2024. | |

Solution to Q2 : Arjun Organics Pvt Ltd | | | | | | | | | |

INCOME STATEMENT AND BALANCE SHEET PREPARATION for the period 30.09.2024 | | | | | | | | | |

Arjun Organics Pvt Ltd | Amount (Rs.) | Amount (Rs.) | | | | | | | |

Revenue From Operations (cash and credit sales)=150000+220000 | | 370000 | | | | | | | |

CLOSING STOCK | | 15000 | | | | | | | |

| | 385000 | | | | | | | |

OPENING STOCK | | 40000 | | | | | | | |

PURCHASES =30000+40000 | | 70000 | | | | | | | |

Gross profit | | 275000 | | | | | | | |

Expenses | | | | | | | | | |

Rent expenses | | 18000 | | | | | | | |

Salaries exp | | 72000 | | | | | | | |

ebitda | | 185000 | | | | | | | |

Depreciation expenses | | 300 | | | | | | | |

EBIT | | 184700 | | | | | | | |

INT | | 0 | | | | | | | |

TAX | | 0 | | | | | | | |

PAT | | 184700 | | | | | | | |

Balance Sheet as on 30.09.2024 | Amount (Rs.) | Amount (Rs.) | | | | | | | |

Equity & Liabilities | | | | | | | | | |

Equity | | | | | | | | | |

Share capital | 200000 | | | | | | | |

Retained earnings | 184700 | | | | | | | |

Total Shareholders' Funds | | 384700 | | | | | | | |

Liabilities | | | | | | | | | |

Non-Current Liabilities – Bank Loan | | 50000 | | | | | | | |

Current Liabilities | | | | | | | | |

Creditors/Trade payables | | 30000 | | | | | | | |

Total Liabilities | | 464700 | | | | | | | |

Assets | | | | | | | | |

Non-Current Assets | | | | | | | | |

Shelves | 12000 | | | | | | | |

(-) Accumulated depreciation on Shelves | 300 | 11700 | | | | | | | |

Refrigerator | | 70000 | | | | | | | |

Total Non-Current Assets | | 81700 | | | | | | | |

Current Assets | | | | | | | | |

Inventories/Closing Stock (note 2) | 15000 | | | | | | | |

Debtors/Trade receivables | 50000 | | | | | | | |

Cash (note 5) | 312000 | | | | | | | |

Prepaid salary | 6000 | | | | | | | |

Total Current Assets | | 383000 | | | | | | | |

Total Assets | | 464700 | | | | | | | |

Notes: | | | | | | | | |

COGS=OPSTK+PURCHASES-CLOSING STOCK - GOODS SPOILT | 40000+70000-8000-15000 | | #REF! | | | |
| | | |

Depreciation on shelves @5% p.a. = $5\% \times 12000 \times 6/12$ = Rs. 300 | | | | | | | | |

Cash Receipt and Payments | | | | | | | | |

| MARKS | | | | | | | | |

Receipts | | | Payments | | | | | | |

CASH IN HAND | 128000 | | Purchases | 40000 | | | | | |

Cash sales =150000+220000-50000 | 320000 | | Rent expenses | 18000 | | | | | |

| | | Salaries exp | 72000 | | | | | |

| | | PREPAID SALARY | 6000 | | | | | |

| | | Cash Balance | 312000 | | | | | |

| 448000 | | | 448000 | | | | | |

c | | | | | | | | |

The estimate of salary that Arjun's sister should be getting cannot form part of the P&L account. Unless Arjun signs a contract with sister and quantifies the exact amount to be paid for it, her salary cannot form part of the financial statements. |||||

Since the friend has given an interest free loan for the first year, the interest on it will not be charged to the P&L account |||||

Money Measurement Concept | Only something that can be measured in terms of money forms part of the Financial statements |||||

d |||||

O/S revenue is shown as an income for the year ending 30th sep 2024 |||||

Revenue realization concept | accrual concept |||||

Revenue is realized when the services are rendered, not when the cash is received. Since services have been rendered it is a revenue for the year ending 30th sep 2024 |||||